

A Evidence chain

1. Governance Exact patient-level harmonization and correction of the 2,205-feature species matrix.
2. Prediction Species-only out-of-domain discrimination remained close to chance with poor calibration.
3. Validation hierarchy Neither locked holdout estimate exceeded its domain-preserving null; the LODO-family contrast was concentrated in the Lee family.
4. Directed transfer Mean AUC was 0.489. Observed reciprocal differences did not exceed the global structured null and were specification-sensitive.
5. Association layers No species or community pathway survived false-discovery-rate correction.
Final interpretation Limited and uncertain cross-cohort transportability; no clinically transportable species-level predictor was supported.

C Potential contributors to limited transportability

Clinical Disease burden, treatment regimen, previous therapy, medications and endpoint definitions.
Ecological Diet, geography, baseline community structure, strain composition and microbial interactions.
Technical Collection, storage, extraction, sequencing depth, profiling database and pathway quantification.
Statistical Small domains, prevalence imbalance, feature sparsity and source-dependent model instability.
Interpretive boundary The present analyses did not assign limited transferability to any single contributor.

B Claim boundary matrix

Supported Limited transportability; sensitivity to validation structure; descriptive reciprocal differences; exploratory taxa; species-pathway ecological correspondence.
Not supported A universal biomarker; a general publication-family effect; a stable direction-dependent biological signal; a causal microbial mechanism; a universal functional pathway.
Required wording Nominal, exploratory, descriptive, context-dependent and processing-system-specific where applicable.

D Requirements for future transportable biomarkers

Independent validation Respect publication, recruitment and processing dependencies; jointly withhold related sites when present.
Standardization Prospectively harmonize stool collection, sequencing, outcome timing and medication exposure.
Bidirectional robustness Evaluate both transfer directions and stability across model complexity and source sample size.
Multilayer confirmation Seek agreement across species, strain, pathway, metatranscriptome, metabolite and immune measurements.
Clinical utility Report calibration and decision benefit with predefined thresholds and prospective replication.